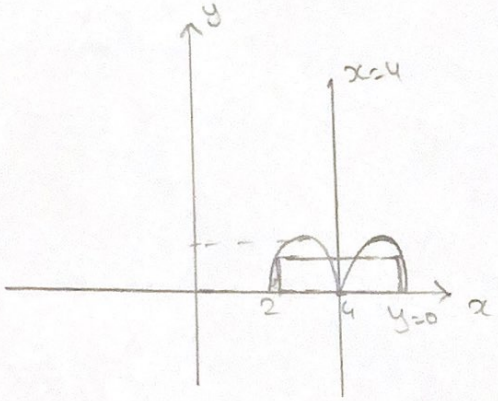
Quiz 6:-

C3: $y = 6x - x^2 - 8$, $y = 0$, about $x = 4$

First quadrant.



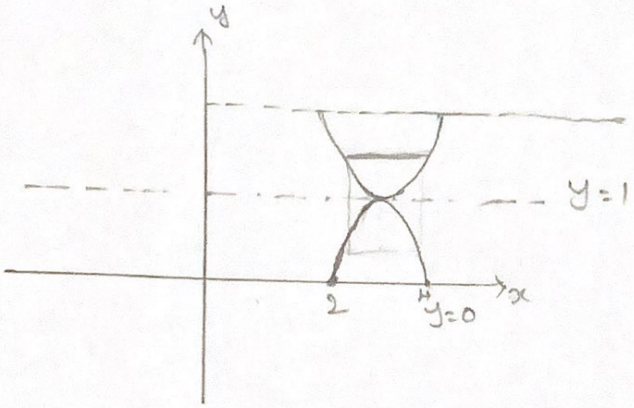
$$V = 2\pi \int_a^b (\text{Radius of the shell})\,(\text{Top} - \text{bottom})\,dx$$

$$V = 2\pi \int_2^4 (4-x)(6x - x^2 - 8)\,dx$$

$$V = 2\pi \int_2^4 (24x - 6x^2 - 4x^2 + x^3 - 32 + 8x)\,dx$$

$$\boxed{V = 2\pi \int_2^4 (32x - 10x^2 + x^3 - 32)\,dx}$$

CH: $y = 6x - x^2 - 8$ and $y = 0$ about $y = 1$



Shell method:

$$V = 2\pi \int_a^b (\text{Radius})(\text{Top} - \text{bottom})\, dx$$

$$V = 2\pi \int_2^4 (4-x)\left(1-(6x-x^2-8)\right) dx$$

$$V = 2\pi \int_2^4 (4-x)\left(1-6x+x^2+8\right) dx$$

$$V = 2\pi \int_2^4 (4-x)(9-6x+x^2)\, dx$$

$$V = 2\pi \int_2^4 (36-9x-24x+6x^2+4x^2-x^3)\, dx$$

$$\boxed{V = 2\pi \int_2^4 (36-33x+10x^2-x^3)\, dx}$$